

Chapter 6: Geometrical figures

Drill practice and mixed revision of all 10 lessons · 82
questions

NSW syllabus outcome

Stage 4 Measurement and space: MA4-ANG-C-01 (angle
relationships, triangles and quadrilaterals) and transformations
of shapes on the Cartesian plane and grids.

NAME

CLASS

CHAPTER

6



Homework. Part 1 is **drill practice**: quick questions on the main skill of each lesson in Chapter 6. Write each answer on its line. Part 2 is **mixed revision**, from **Easy** to **Challenging**. Each question is tagged with its lesson, so you can look back at that lesson's summary if you get stuck.

YOU DO On your own: Part 1, drill practice.

QUESTIONS 1–30

6.01 Transforming shapes**Easy** Translate (slide) the point. Write where it ends up.**1** $(-5, 0)$: 1 right, 2 up**2** $(1, 0)$: 4 left**3** $(-1, 4)$: 6 left, 2 up**Medium** Rotate (turn) the point about the origin.**4** $(5, -2)$, 90° clockwise**5** $(4, -1)$, 180° **6** $(-2, 4)$, 90° clockwise**6.02 Combined transformations****Easy** Name the transformation: translation, reflection or rotation.**7** a lift going up**8** a mirror image**Medium** Do both transformations in order. Where does the point end up?**9** $(4, 1)$: 1 right, then reflect in the x-axis**10** $(4, -1)$: 3 right, then reflect in the x-axis**6.03 Line symmetry****Easy** How many axes (lines) of symmetry?**11** regular octagon**12** semicircle**13** isosceles trapezium**Medium** How many axes of symmetry does a regular polygon with this many sides have?**14** 5 sides**15** 13 sides**16** 4 sides**17** 15 sides**6.04 Rotational symmetry****Easy** Order of rotational symmetry (write 1 for none).**18** kite**19** scalene triangle**20** square**Medium** A regular polygon has this many sides. Write its order of rotational symmetry and the smallest angle of turn.**21** 5 sides**22** 9 sides**23** 24 sides**6.05 Classifying triangles****Easy** Classify by angles: acute, right or obtuse.**24** 110° , 41° , 29° **25** 81° , 34° , 65° **26** 65° , 81° , 34° **27** 29° , 129° , 22° **Medium** Classify by sides and by angles.**28** angles 60° , 60° , 60° **29** angles 90° , 45° , 45° **30** angles 130° , 31° , 19°

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 31–57

6.06 Angle sum of a triangle

Easy Two angles of a triangle are given. Find the third. (Angles add to 180° .)

31 58° and 66°

32 69° and 76°

33 52° and 113°

34 58° and 62°

Medium Isosceles triangle: find the missing angles.

35 base angle 28°

36 base angle 41°

6.07 Exterior angle of a triangle

Easy The two interior opposite angles are given. Find the exterior angle (add them).

37 61° and 61°

38 82° and 16°

39 58° and 28°

40 55° and 100°

Medium The exterior angle and one interior opposite angle are given. Find the other interior opposite angle.

41 exterior 74° , 23°

42 exterior 170° , 102°

6.08 Quadrilaterals

Easy Name the quadrilateral.

43 diagonals cross at 90° and one diagonal bisects the other

44 opposite angles equal, no right angles, sides not all equal

Medium How many pairs of parallel sides? How many right angles?

45 parallelogram

46 isosceles trapezium

47 square

6.09 Angle sum of a quadrilateral

Easy Three angles of a quadrilateral are given. Find the fourth. (Angles add to 360° .)

48 87° , 106° , 96°

49 76° , 120° , 111°

50 63° , 75° , 123°

Medium A kite has two equal opposite angles, x . Find x .

51 other angles 123° and 67°

52 other angles 74° and 54°

53 other angles 62° and 80°

6.10 Properties of quadrilaterals

Easy True or false?

54 Co-interior angles in a parallelogram add to 180° .

55 A kite has two pairs of equal opposite sides.

Medium Use the properties to find the missing value.

56 square: perimeter 48 cm. The side?

57 isosceles trapezium: one base angle is 65° . The other base angle?

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 58–67

58

Easy

A point at (3, 2) is translated. Where is it now?
2 left, 4 up Lesson 6.01

59

Easy

Give the order of rotational symmetry.
rhombus Lesson 6.04

60

Easy

The interior opposite angles are given. Find the exterior angle.
47° and 68° Lesson 6.07

61

Easy

True or false?
A rectangle is a parallelogram. Lesson 6.10

62

Easy

How many axes of symmetry?
rhombus Lesson 6.03

63

Easy

Two angles of a triangle are given. Find the third.
38° and 97° Lesson 6.06

64

Easy

Three angles of a quadrilateral are given. Find the fourth.
110°, 75°, 105° Lesson 6.09

65

Easy

Name the transformation.
a slide 4 units up Lesson 6.02

66

Easy

Classify the triangle by its angles.
25°, 35°, 120° Lesson 6.05

67

Easy

Name the quadrilateral.
a rectangle with 4 equal sides Lesson 6.08

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 68–77

- 68** **Medium** A point is at (2, 3) on a grid. Where is it after a translation 4 right and 1 down? Lesson 6.01

- 70** **Medium** The exterior angle is 125° and one interior opposite angle is 60° . Find the other. Lesson 6.07

- 72** **Medium** How many axes of symmetry does a kite have? Lesson 6.03

- 74** **Medium** Three angles are 100° , 85° and 95° . Find the fourth. Lesson 6.09

- 76** **Medium** Classify a triangle with angles 90° , 45° and 45° . Lesson 6.05

- 69** **Medium** Does a kite have rotational symmetry? Lesson 6.04

- 71** **Medium** State one property of the diagonals of a rhombus. Lesson 6.10

- 73** **Medium** A right-angled triangle has an angle of 28° . Find the other angle. Lesson 6.06

- 75** **Medium** A point (1, 2) is translated 3 right then 4 up. Where is it? Lesson 6.02

- 77** **Medium** Name a quadrilateral with exactly one pair of parallel sides. Lesson 6.08

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 78–82

- 78** **Challenging** Describe what a 90° clockwise rotation does to an arrow pointing up. Lesson 6.01

- 80** **Challenging** How many axes of symmetry does a regular hexagon have? Lesson 6.03

- 82** **Challenging** Classify a triangle with angles 110° , 35° and 35° by sides and angles. Lesson 6.05

- 79** **Challenging** The exterior angle is 140° . Find the adjacent interior angle. Lesson 6.07

- 81** **Challenging** A parallelogram has an angle of 65° . Find the other angles. Lesson 6.09